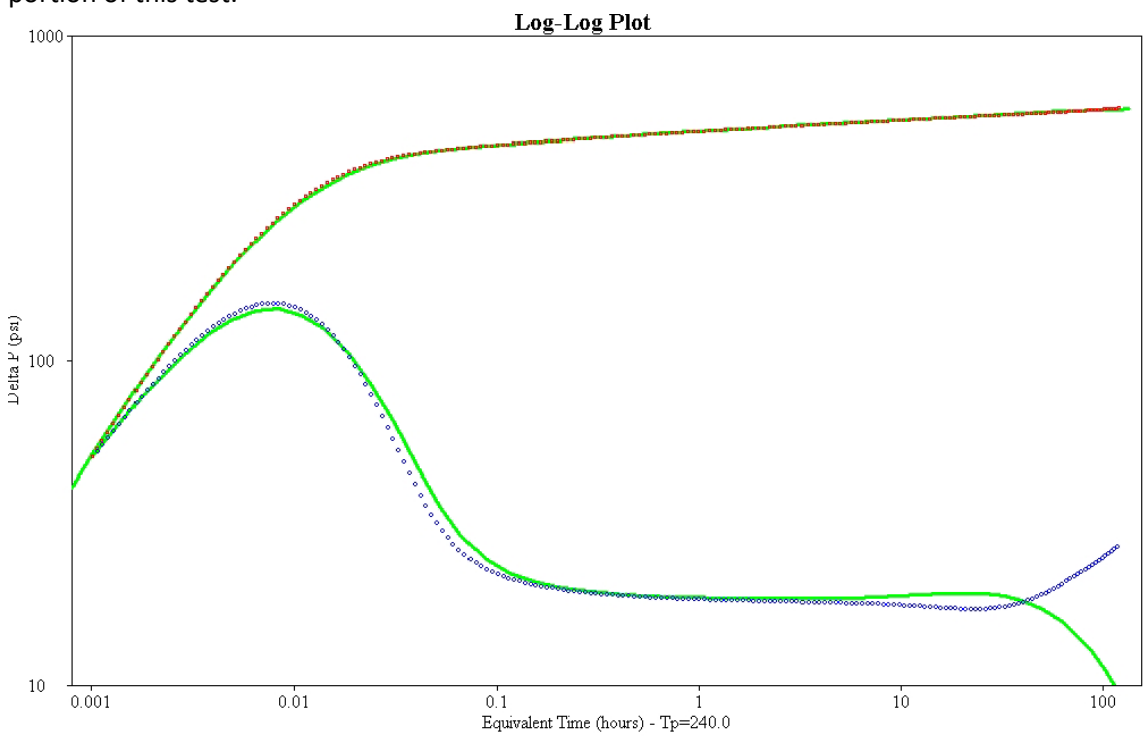
PGE 379/383 Well Testing
Midterm
Spring 2026

This is a take home exam. Open book and open notes. Your exam with answers and screenshots as appropriate has to be returned in Canvas by 9 am, Monday, March 30th. Do not collaborate with your classmates to work this exam.

The plot below is a standard diagnostic log-log plot that is used in well testing. This particular presentation has two lower and two upper curves showing different responses. This well has flowed for 24000 hours and then was shut-in for 120 hours. This is diagnostic for the build-up portion of this test.

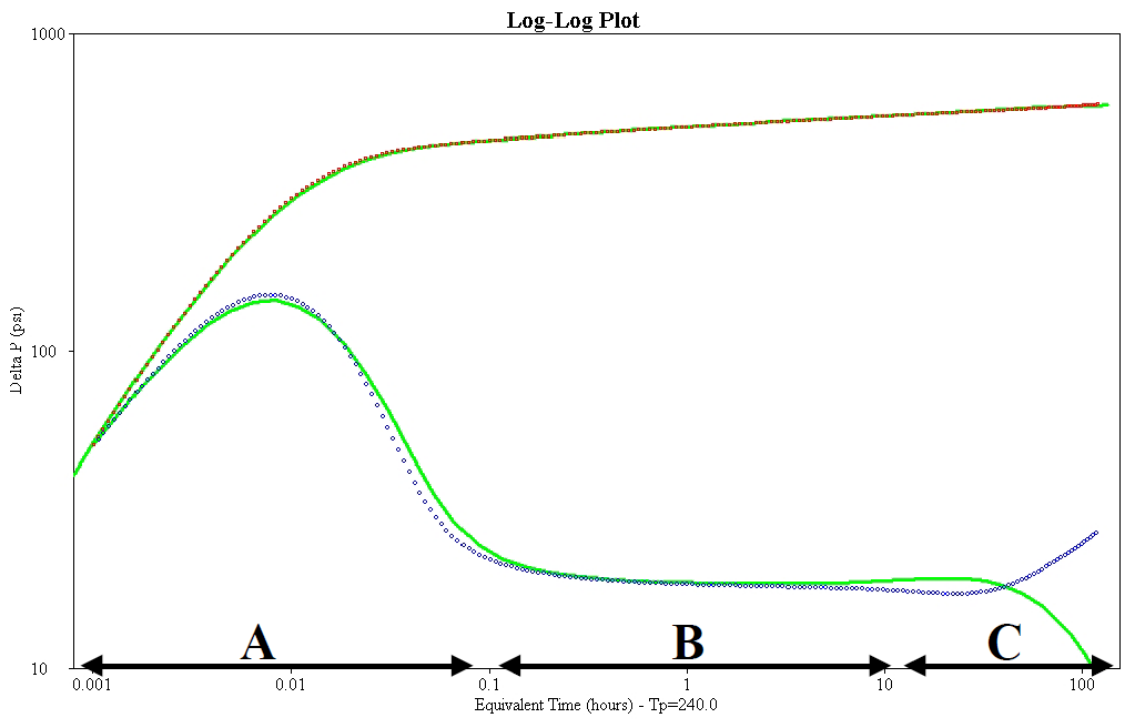


Choose the best answer:

1. Each point on the upper curve represents:
    a. The difference in pressure between the initial pressure and the current pressure.
    b. The difference in pressure between the current pressure and the last flowing pressure.
    c. Neither a nor b
2. Each point on the lower curves represents:
    a. The derivative of current pressure with respect to time that the well has been shut-in
    b. The derivative of current pressure w.r.t. log time that the well has been shut in

c. The derivative of current pressure w.r.t. square root of time that the well has been shut-in
d. The derivative of current pressure w.r.t. inverse square root of time that the well has been shut-in.

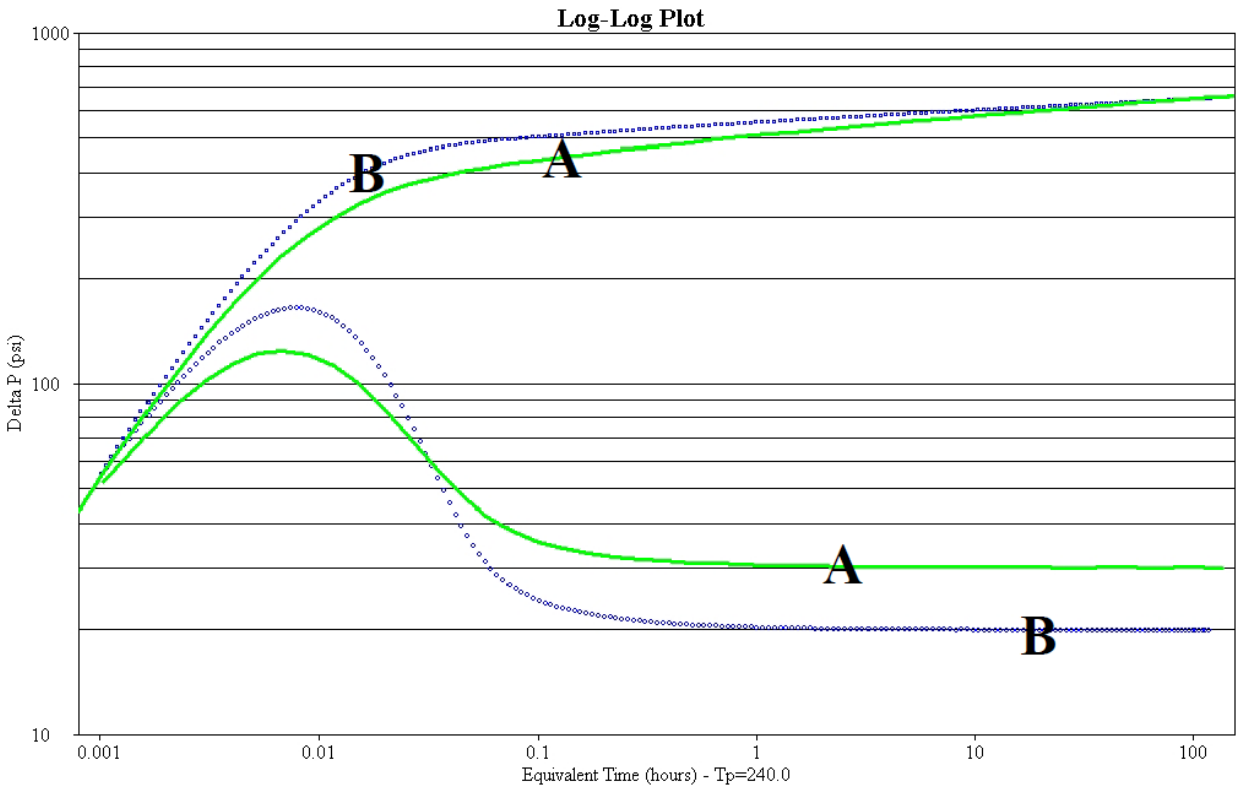
This is the same diagnostic plot as shown in questions 1 and 2. There are three time periods; A, between 0.001 and 0.1 hours, B, between 0.1 and 10, and C after 10 hours.



Choose the best answer

3. Which flow regime best characterizes period A?
    a. Boundary effects
    b. Linear flow
    c. Radial Flow
    d. Spherical Flow
    e. Wellbore Storage or after flow
4. Which flow regime best characterizes period B?
    a. Boundary effects
    b. Linear flow
    c. Radial Flow
    d. Spherical Flow
    e. Wellbore Storage or after flow
5. Which flow regime best characterizes period C?
    a. Boundary effects
    b. Linear flow
    c. Radial Flow
    d. Spherical Flow
    e. Wellbore Storage or after flow

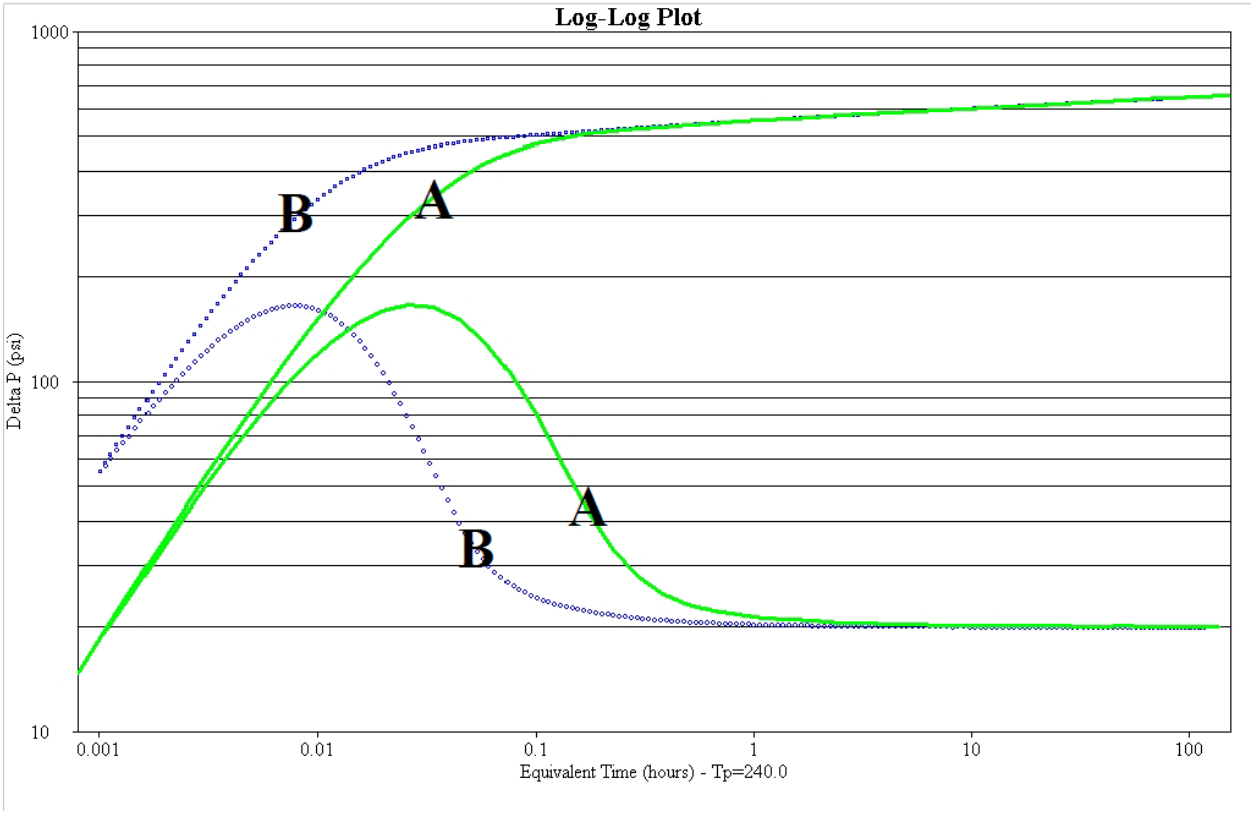
The plot below shows diagnostic plots for two wells. They are identical in most aspects – their PVT, net pay thickness, porosities, saturations, compressibility and even flow rate histories.



6. Which of the following statements best summarizes what you can observe from this diagnostic plot?
    a. **A** has a higher perm and the same skin as **B**.
    b. **A** has a lower perm and the same skin as **B**.
    c. **A** has a higher perm and a lower skin than **B**.
    d. **A** has a lower perm and a lower skin than **B**.
    e. **A** has a higher perm and a higher skin than **B**.
    f. **A** has a lower perm and a higher skin than **B**.

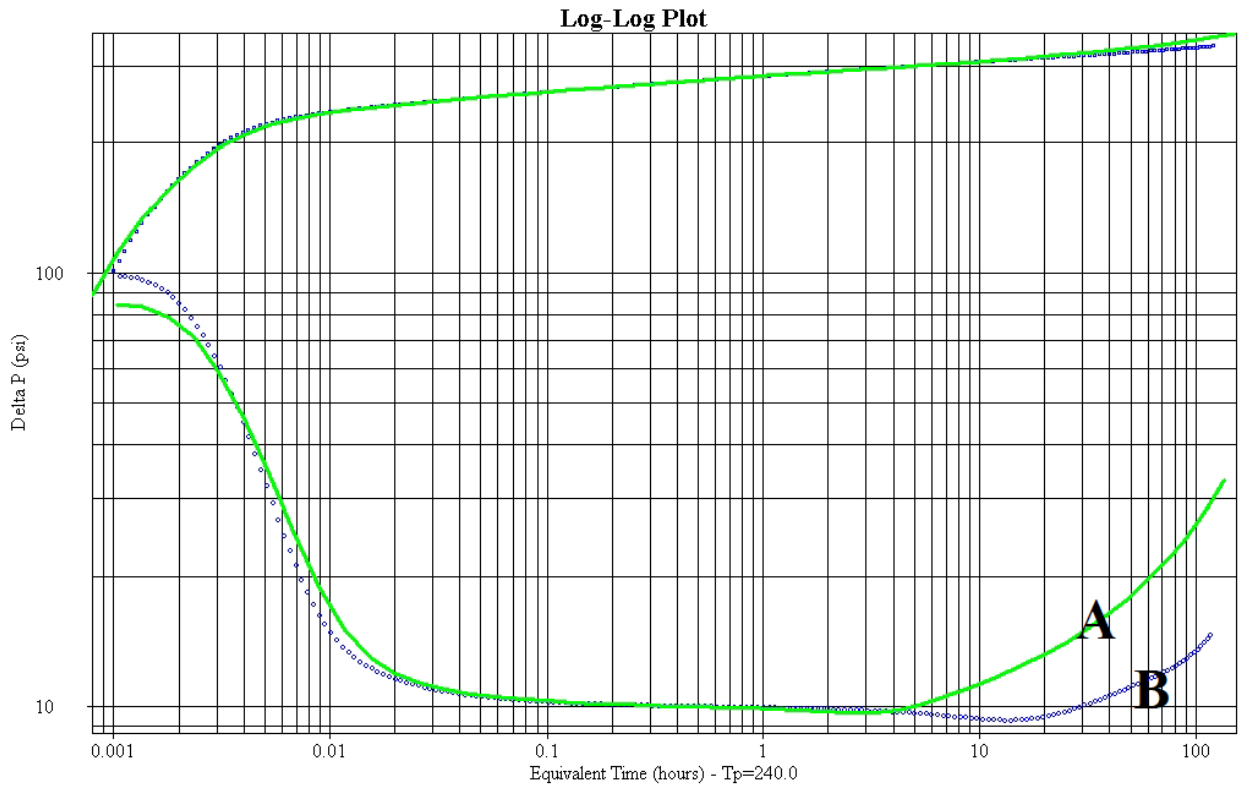
Extra Credit (+5 points). What is the ratio of perm of well A to perm of well B?

This figure shows the diagnostic plots for two wells. They are identical in most aspects – their PVT, net pay thickness, porosities, saturations, compressibility and even flow rate histories.



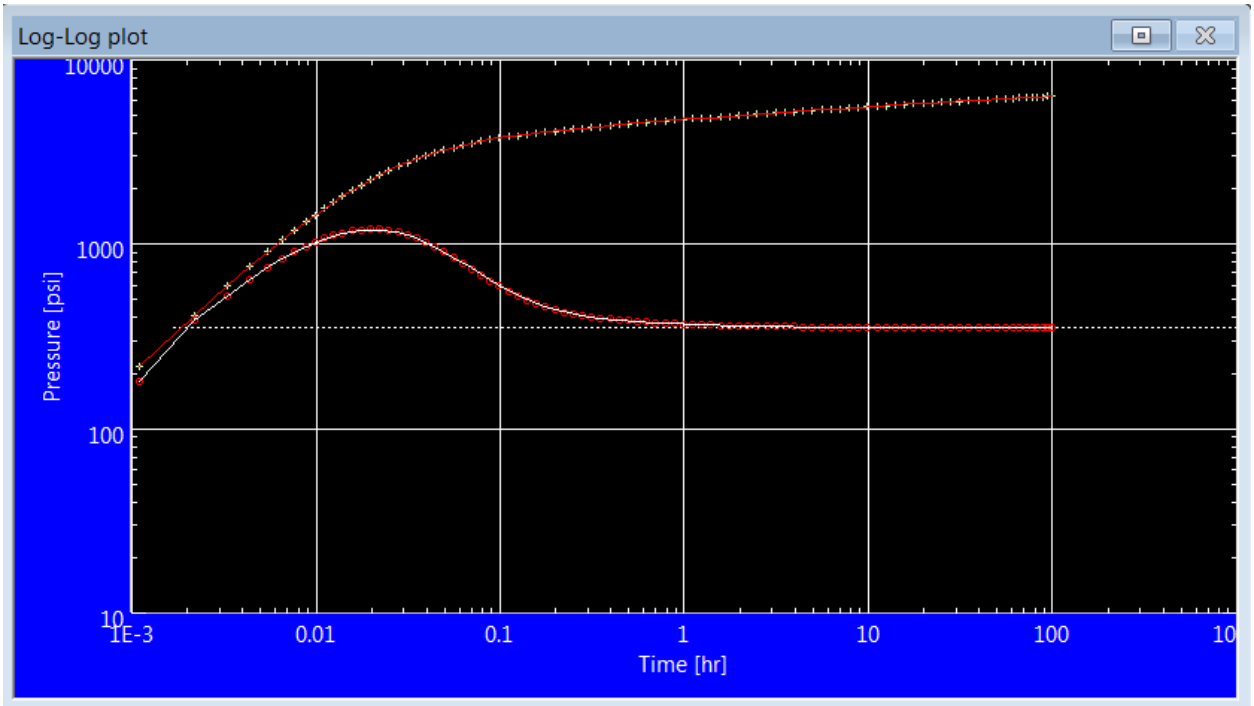
7. Which one of the flowing statements best summarizes what you can observe from the diagnostic plot?
    a. **A** has a higher wellbore storage and the same perm as **B**.
    b. **A** has a lower wellbore storage and the same perm as **B**.
    c. **A** has a higher wellbore storage and a lower perm than **B**.
    d. **A** has a lower wellbore storage and a lower perm than **B**.
    e. **A** has a higher wellbore storage and a higher perm than **B**.
    f. **A** has a lower wellbore storage and a higher perm than **B**.

This figure shows the diagnostic plots for two wells. They are identical in most aspects – their PVT, net pay thickness, porosities, saturations, compressibility and even flow rate histories.



8. Which one of the following statements best summarizes what you can observe from the diagnostic plot?
    a. They have the same perm, the skin of **A** is higher than **B** and the boundary effects in **A** are closer than in **B**.
    b. They have the same skin, the boundary effects are the same distance away from both wells, but the perm of **A** is higher than **B**.
    c. They have the same skin, the boundary effects are the same distance away from both wells, but the perm of **B** is higher than **A**.
    d. They have the same skin and perm and the boundary effects in A are closer than in B.
    e. They have the same skin and perm and the boundary effects in B are closer than in A.
    f. They have the same skin and perm and without more work you cannot make any statement about the relative distance of A and B from the well.

9. The log-log diagnostic plot shown below is for a buildup test in a vertical well. Please answer the following questions. Before shut-in, this well was producing at a constant rate of 5000 stb/d. The oil PVT properties are B_o = 1.2 and μ_o = 0.5 cp.



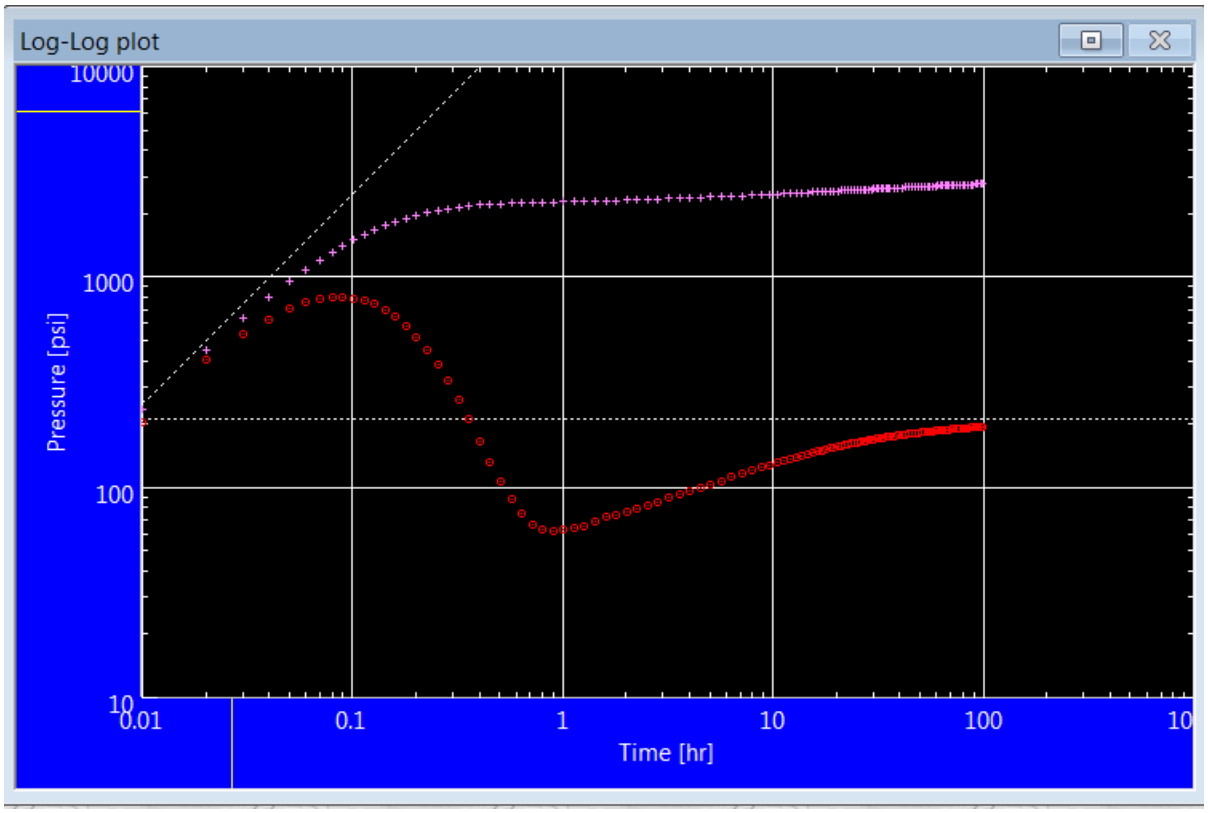
a) What are the 2 curves shown above? Please label each curve.

b) Label and name the different flow regimes present (there are 2).

c) What is the equation of "kh" from the derivative curve?

d) What is the value of "kh" (md-ft) from the above plot?

10. Shown below is a log-log diagnostic plot for a buildup test in a horizontal well. Prior to the well shut-in, the well flowed for 115 hrs at a constant rate of 5000 stb/d. The well was then shut-in for 100 hrs to measure the pressure buildup response. Please answer the following questions.



a. Name and label the different flow regimes present in this plot? (Note that there are 4).
b. From this plot, estimate the far field "kh".

11. Answer the following questions about Hall Plots
    a. What they are used for?
    b. How are they constructed?

12. What are Fall off tests used for? Please describe what they are, what subsurface information you can get from them and what kind of wells they are performed on.

13. Analyze the following double shut-in Drillstem test and determine the following static reservoir and well properties :
    a. Initial reservoir pressure
    b. Formation permeability
    c. Total skin factor
    d. Damage ratio

Initial Flow		Initial Buildup		Final Flow		Final Buildup	
Time	Pressure	Time	Pressure	Time	Pressure	Time	Pressure
min	psig	min	psig	min	psig	min	psig
0	83	0	371	0	411	0	656
5	216	2	982	7	455	2	1003
10	260	4	1020	14	515	4	1025
15	313	6	1041	21	574	6	1040
20	371	8	1056	28	609	8	1051
		10	1068	35	656	10	1059
		12	1077			12	1066
		14	1083			14	1072
		16	1089			16	1077
		18	1094			18	1081
		20	1097			20	1084

<u>Reservoir and well data</u>

Average production rate =	660 stb/d
Formation thickness =	69 ft
Oil viscosity =	2.4 cp
Oil formation volume factor =	1.2 rb/stb
Wellbore radius =	0.33 ft
Porosity =	0.15
Total compressibility =	20×10^{-6} psi^{-1}

Note: for the final buildup, use $t_p = (t_1 + t_2)$ where t_1 and t_2 are the initial and final flow periods, respectively.

14. The shut-in time and pressure table below shows pressure buildup data from an oil well with an estimated drainage radius of 2640 ft. Before shut-in the well had produced at a stabilized rate of 4900 stb/d for 310 hours. Known reservoir data are

Depth = 10476 ft
r_w = 4.25 in
$C_t = 22.6 \times 10^{-6}$ psi^{-1}
q_o = 4900 stb/d
h = 482 ft
p_{wf} (Δt=0) = 2761 psig
μ_o = 0.20 cp
ϕ=0.09
B_o=1.55 rb/stb
Casing ID = 6.276 in
t_p = 310 hours

Δt	p_{ws}
(hours)	(psig)
0	2761
0.1	3057
0.21	3153
0.31	3234
0.52	3249
0.63	3256
0.73	3260
0.84	3263
0.94	3266
1.05	3267
1.15	3268
1.36	3271
1.68	3274
1.99	3276
2.51	3280

3.04	3283
3.46	3286
4.08	3289
5.03	3293
5.97	3297
6.07	3297
7.01	3300
8.06	3303
9.00	3305
10.05	3306
13.09	3310
16.02	3313
20.00	3317
26.07	3320
31.03	3322
34.98	3323
37.54	3323

Based on this data, estimate the following;

a. Formation permeability
b. Skin
c. Formation flow efficiency
d. Time to reach pseudo-steady state, t_{pss}
e. P*
f. Estimate the average reservoir pressure for the drainage area

15. An oil well produced 4410 stb of oil before it was shut in for a pressure buildup test. Before the well was shut-in, the average production rate was 105 stb/d. The pressure and other pertinent data are as follows:

Δt, hrs	P_{ws}, psig
0.0	2665
0.1	2940
0.2	2970
0.5	3009
1.0	3040
2.0	3070
5.0	3111
10.0	3148
20.0	3192
50.0	3258
100.0	3311
200.0	3361

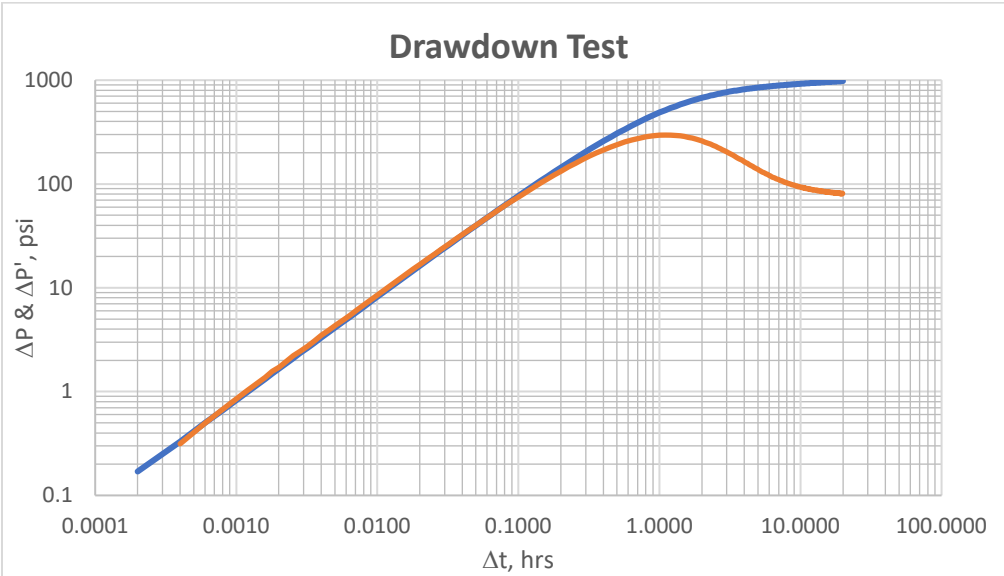
μ= 0.87 cp
B_o = 1.27 rb/stb
C_t = 18.4 x 10^{-6} psi^{-1}
ϕ= 0.16
Well diameter = 7.92 in
h = 16 ft
drainage area, A Is very large

Analyze the pressure data presented above by using Horner's method and estimate the following parameters

a. Formation permeability
b. Total skin factor
c. Average reservoir pressure

d. Any other parameter you think is appropriate in view of the data from the trend then please explain.

16. Refer to the well test data plot below to answer the questions below.



A drawdown test was performed on an oil well for a period of 20 hours. During the test, this well produced at a constant rate of 115 stb/d of oil. Other pertinent data necessary to analyze this test are as follows:

μ_o, cp	1.3
B_o, rb/stb	1.73
r_w, ft	0.25
ϕ	0.186
h, ft	8
q, stb/d	115
c_t, psi^{-1}	1.40E-05
P_i, psia	3700

Questions:

A. Identify the 2 types of lines shown in this figure. Please annotate the plot to show your answers.
B. Up to what approximate time, in hours is this test exhibiting only wellbore storage?
C. What is the approximate "y value" on this plot that this well flowing at infinite acting radial flow conditions (IARF)?
D. Based on the IARF value, what is the reservoir permeability (md), the skin factor and the storage (bbl/psi)?

17. A Horizontal well test.

Draw a typical Δp and Derivative plot for a horizontal well test. Label each of the 5 flow regimes on the derivative curve.